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Discounted Cash Flow (DCF) based methods

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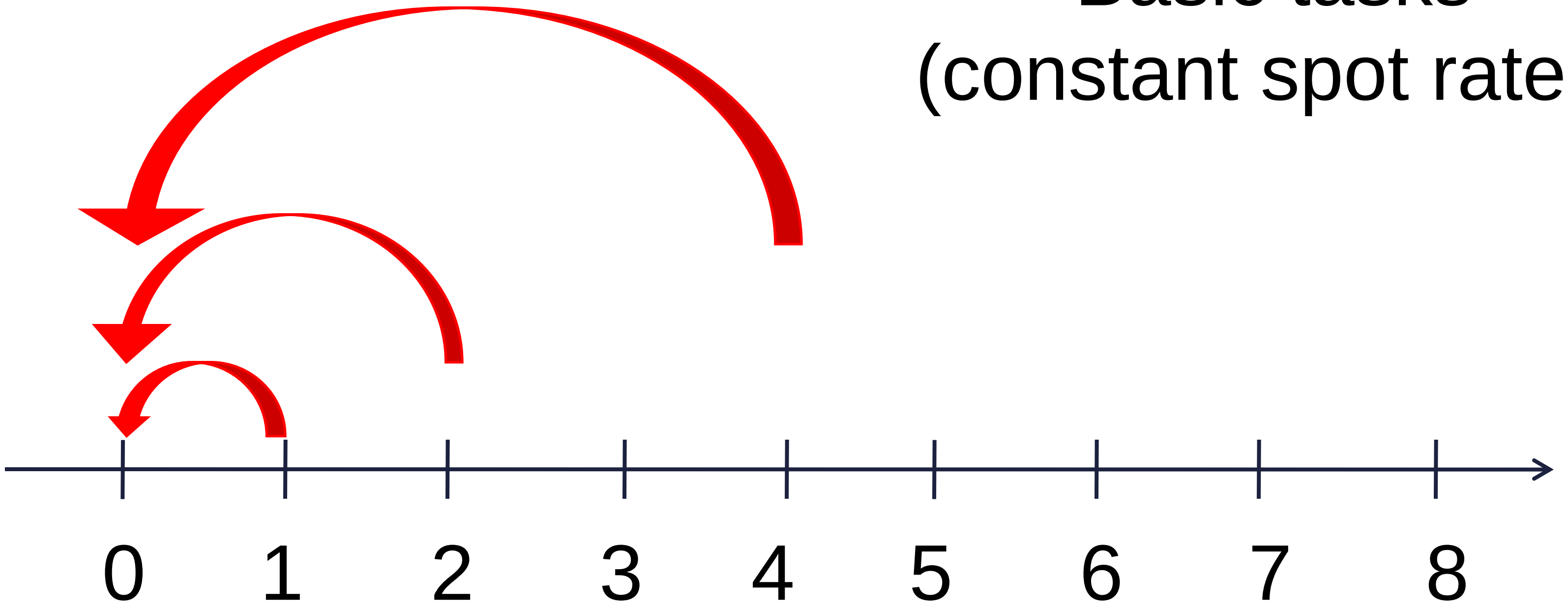
Corporate finance and reality

Corporate finance: flat yield curve, spot rates, required rates given

Reality: variable yield curve, forward rates, required rates are to estimated

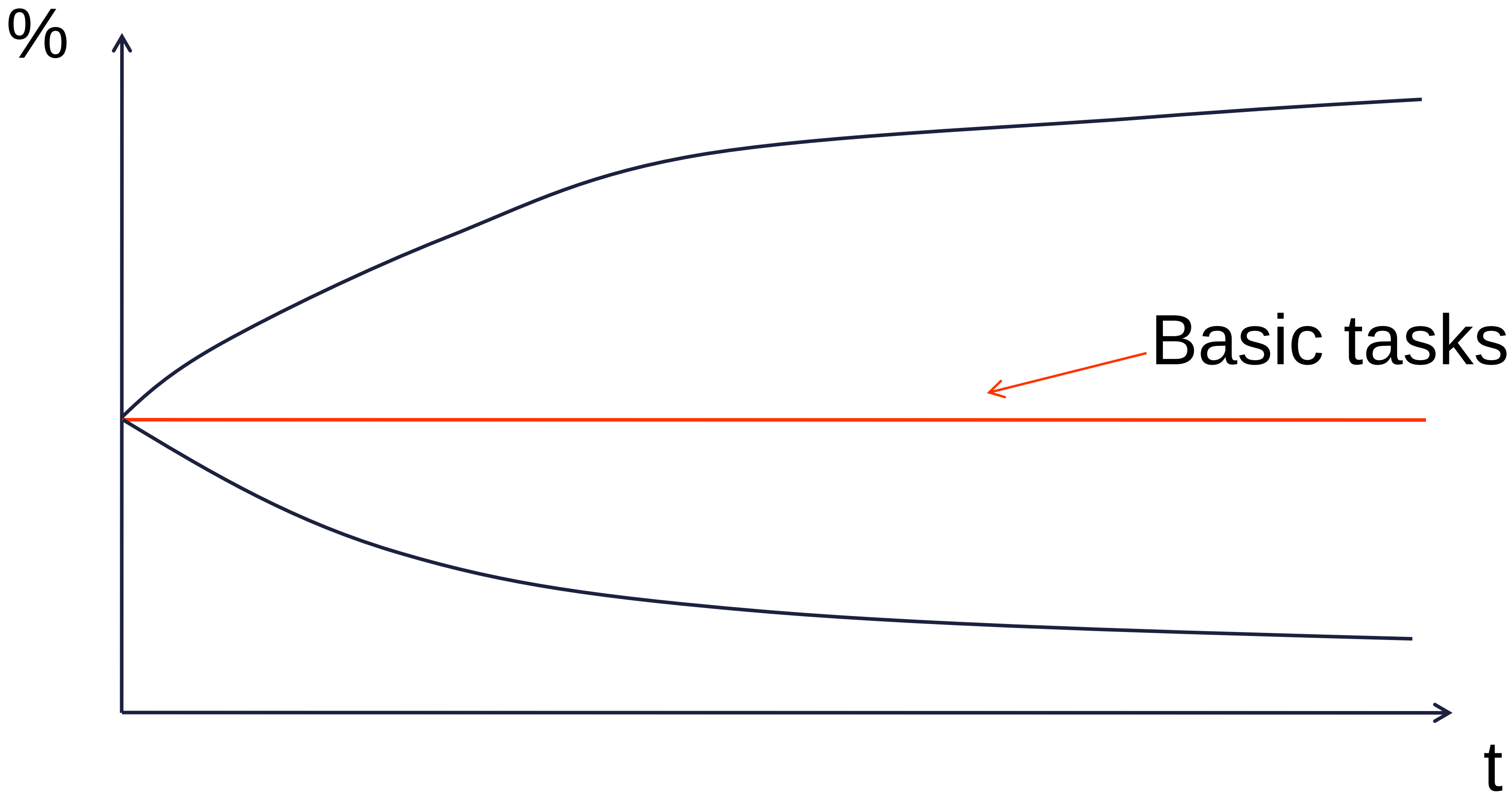
Calculating the present value 1

Basic tasks
(constant spot rates)

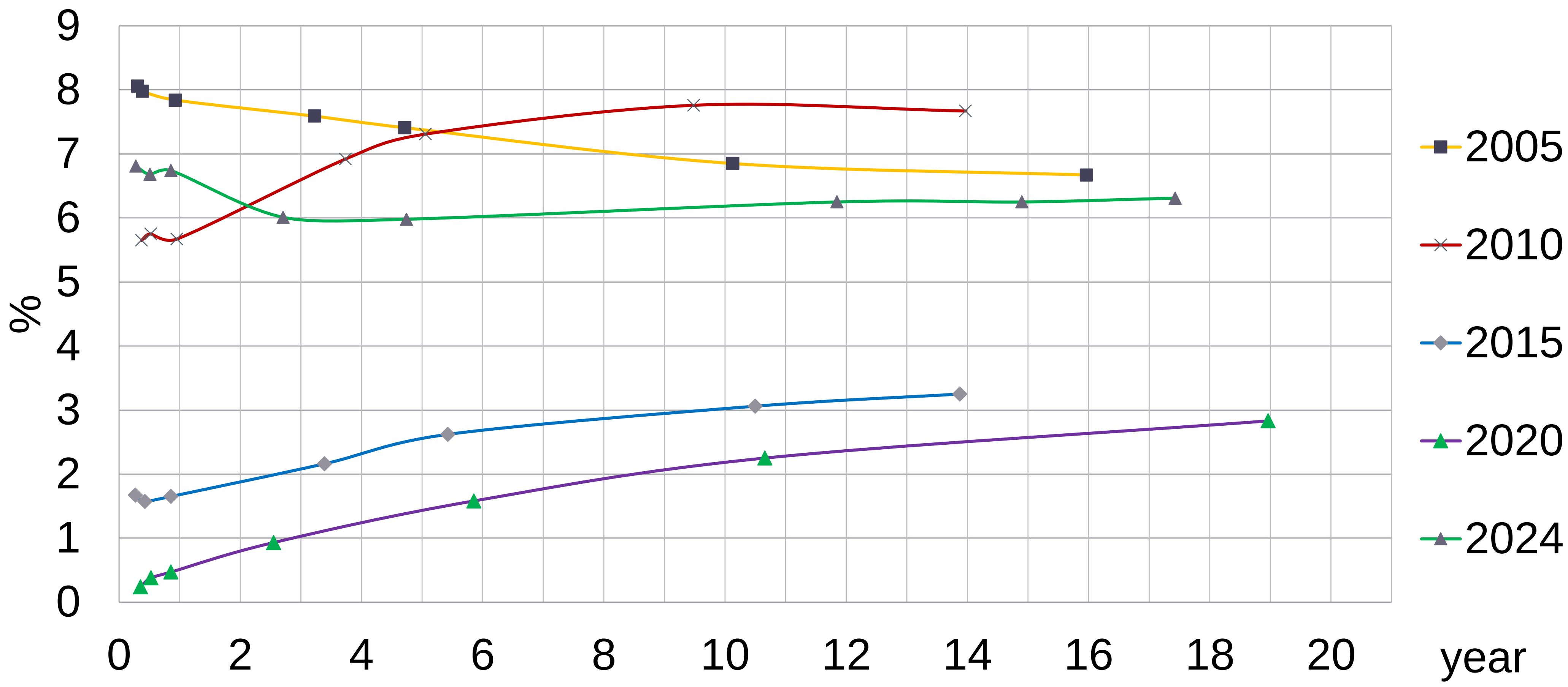


$$\frac{1}{(1+r)} \qquad \frac{1}{(1+r)^2} \qquad \frac{1}{(1+r)^4}$$

The form of the yield curve



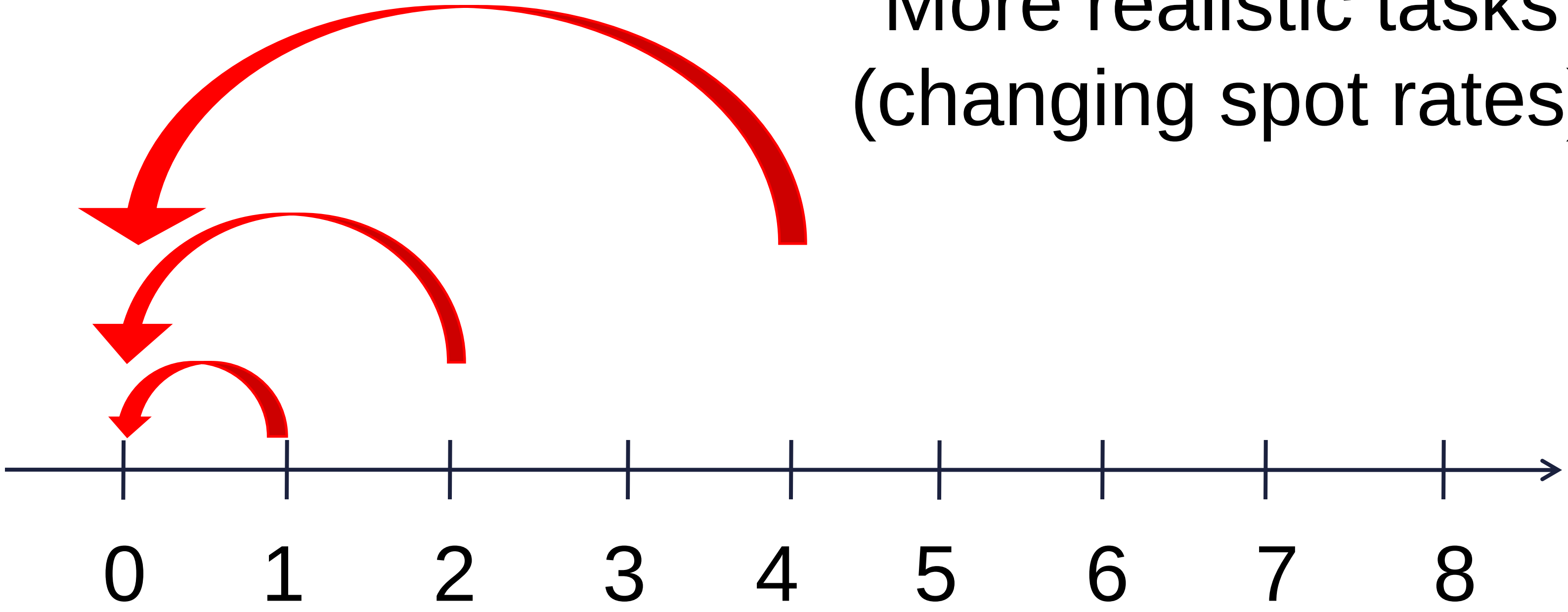
Required spot rates in Hungary (15 February)



Source: www.akk.hu

Calculating the present value 2

More realistic tasks
(changing spot rates)

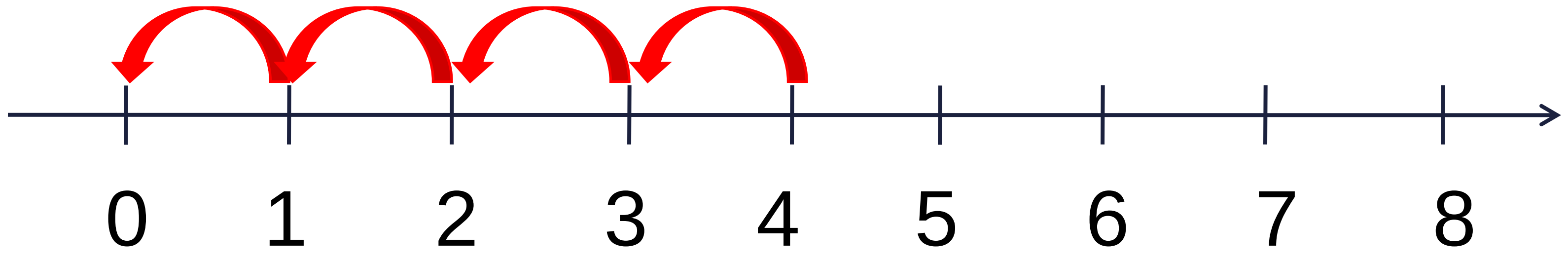


$$\frac{1}{(1+r_1)} \quad \frac{1}{(1+r_2)^2} \quad \frac{1}{(1+r_4)^4}$$

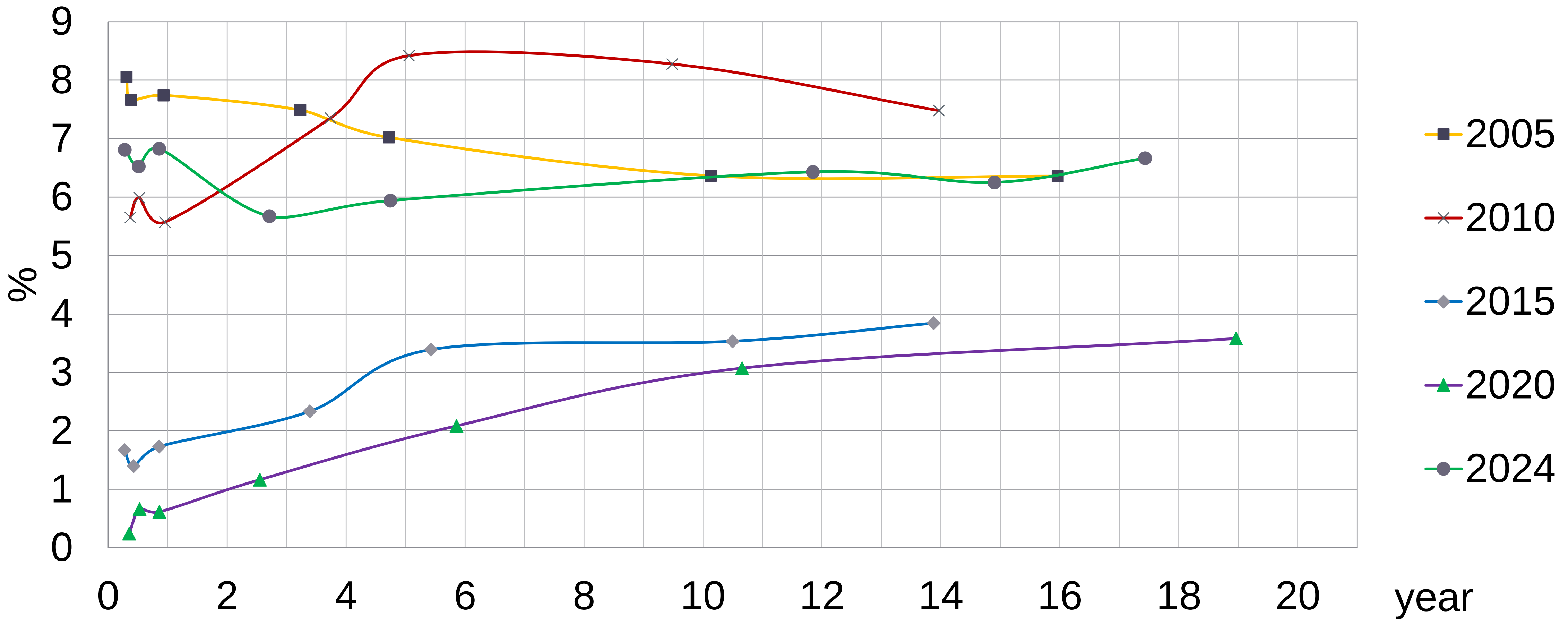
Calculating the present value 3

Reality
(forward rates)

$$\frac{1}{(1+{}_0f_1)} * \frac{1}{(1+{}_1f_2)} * \ldots * \frac{1}{(1+{}_3f_4)}$$



Required 1-year forward rates in Hungary (15 February)



A simple task

Year	0	1	2	3	4
CF		100.00	150.00	200.00	30.00
Spot r		8.00%	9.00%	10.00%	11.00%

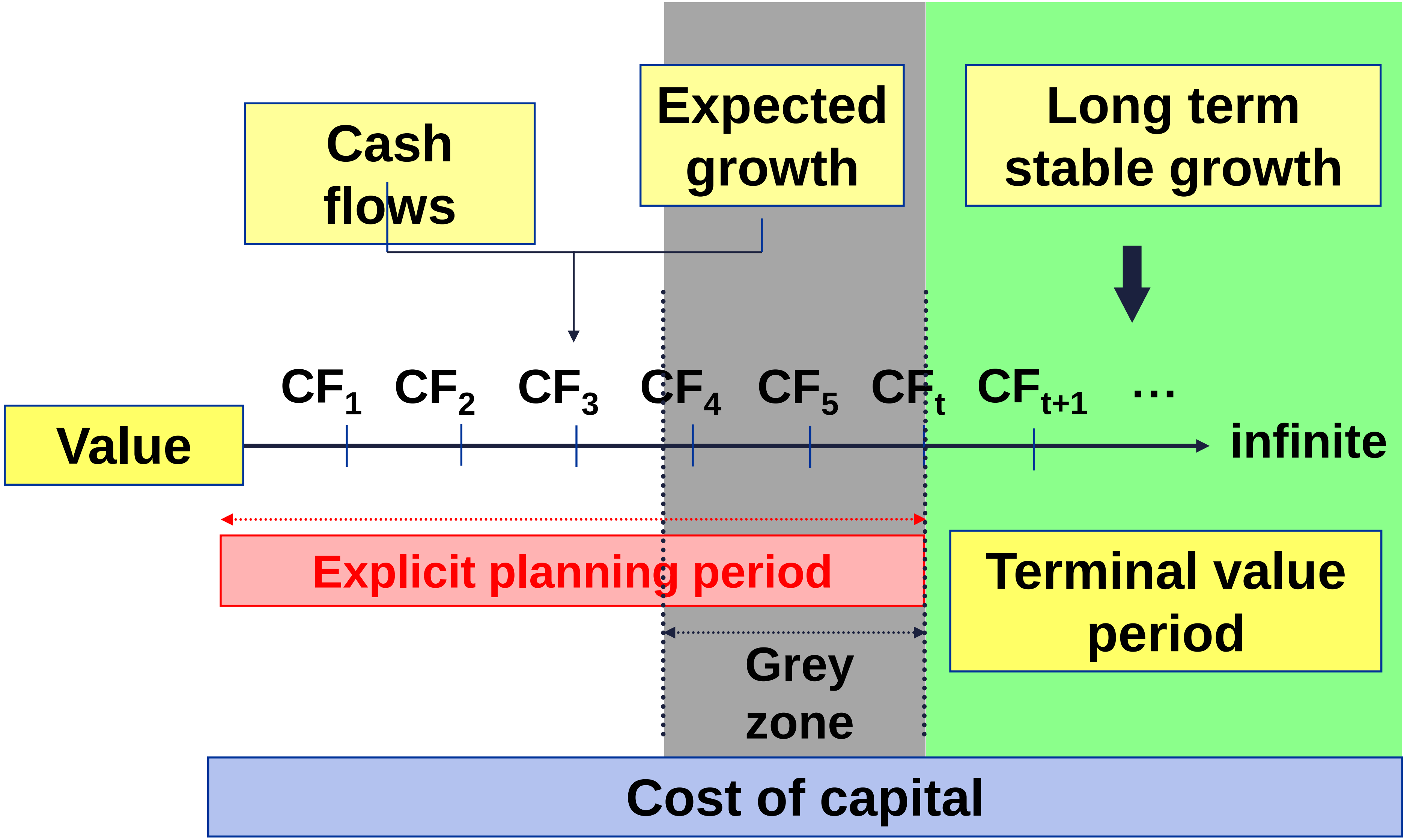
PV		92.59	126.25	150.26	19.76
Sum PV	388.87				

Forward r 1 year		8.00%	10.01%	12.03%	14.05%
Value boy		388.87	319.98	202.01	26.30

The general DCF model 1

$$NPV = \sum_{i=0}^{\infty} \frac{CF_i}{\prod_{j=1}^{i} (1 + r_j)}$$

The general DCF model 2



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Durations

Explicit period

The longer the better, min 5-7 years

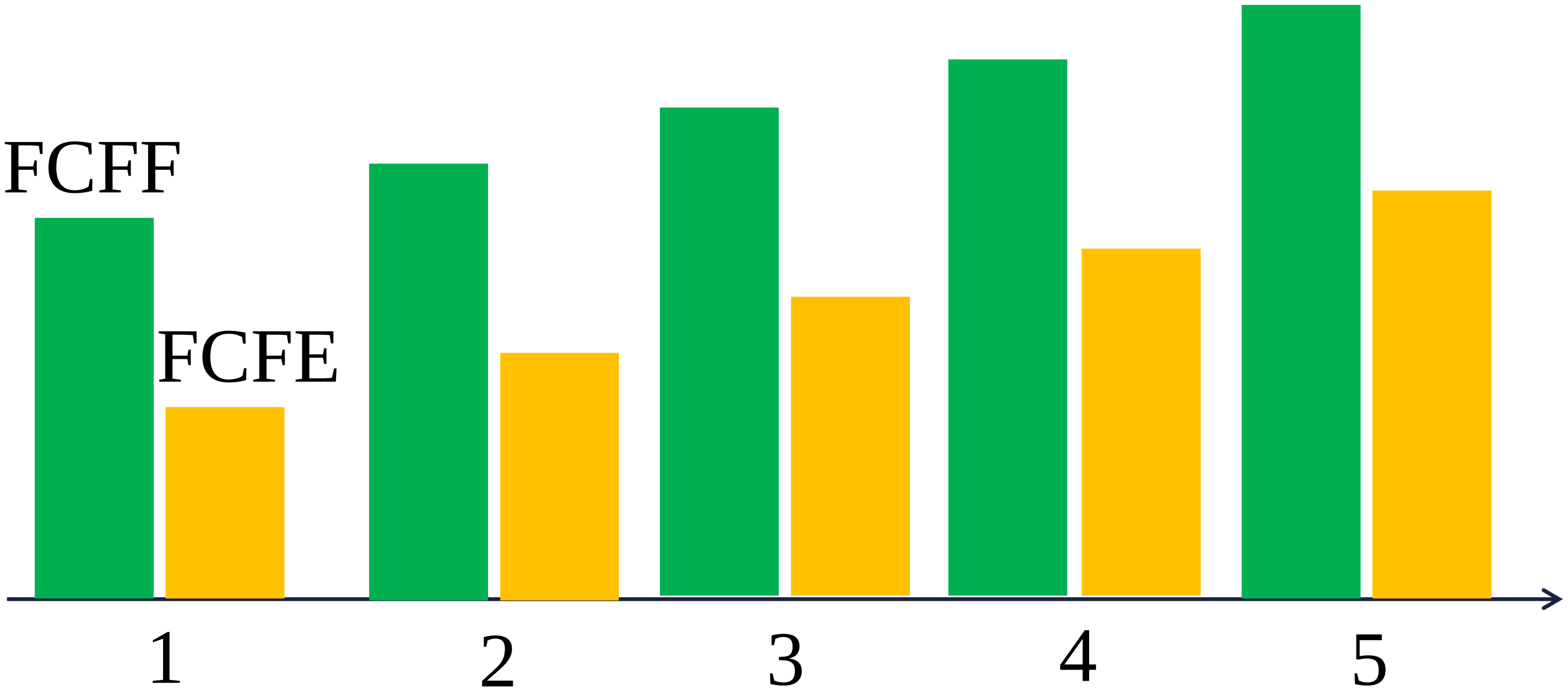
Terminal Value period

Only to start when everything will be stable

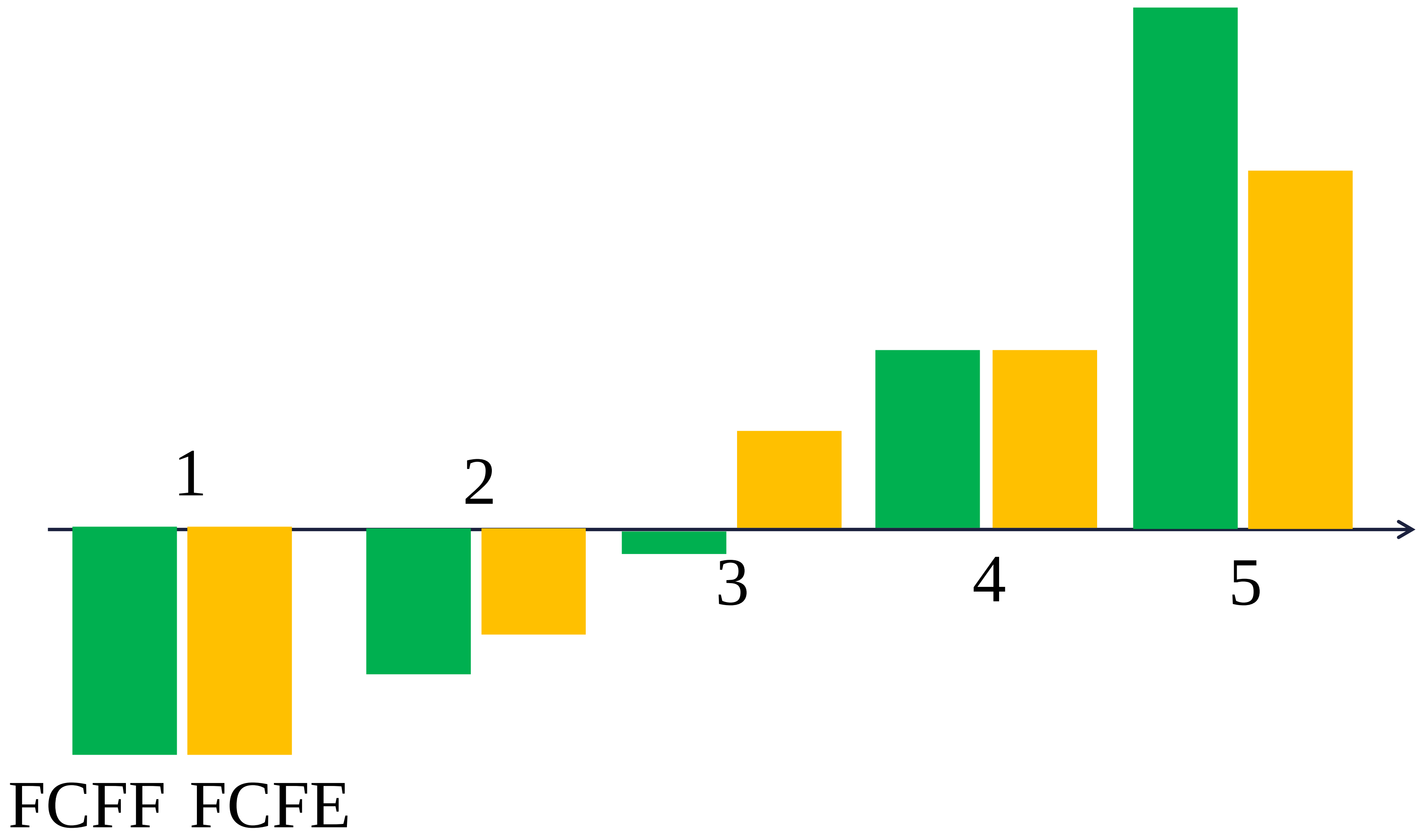
Grey (twilight) zone

In-between the earlier two, may use a simplified (value driver only) model

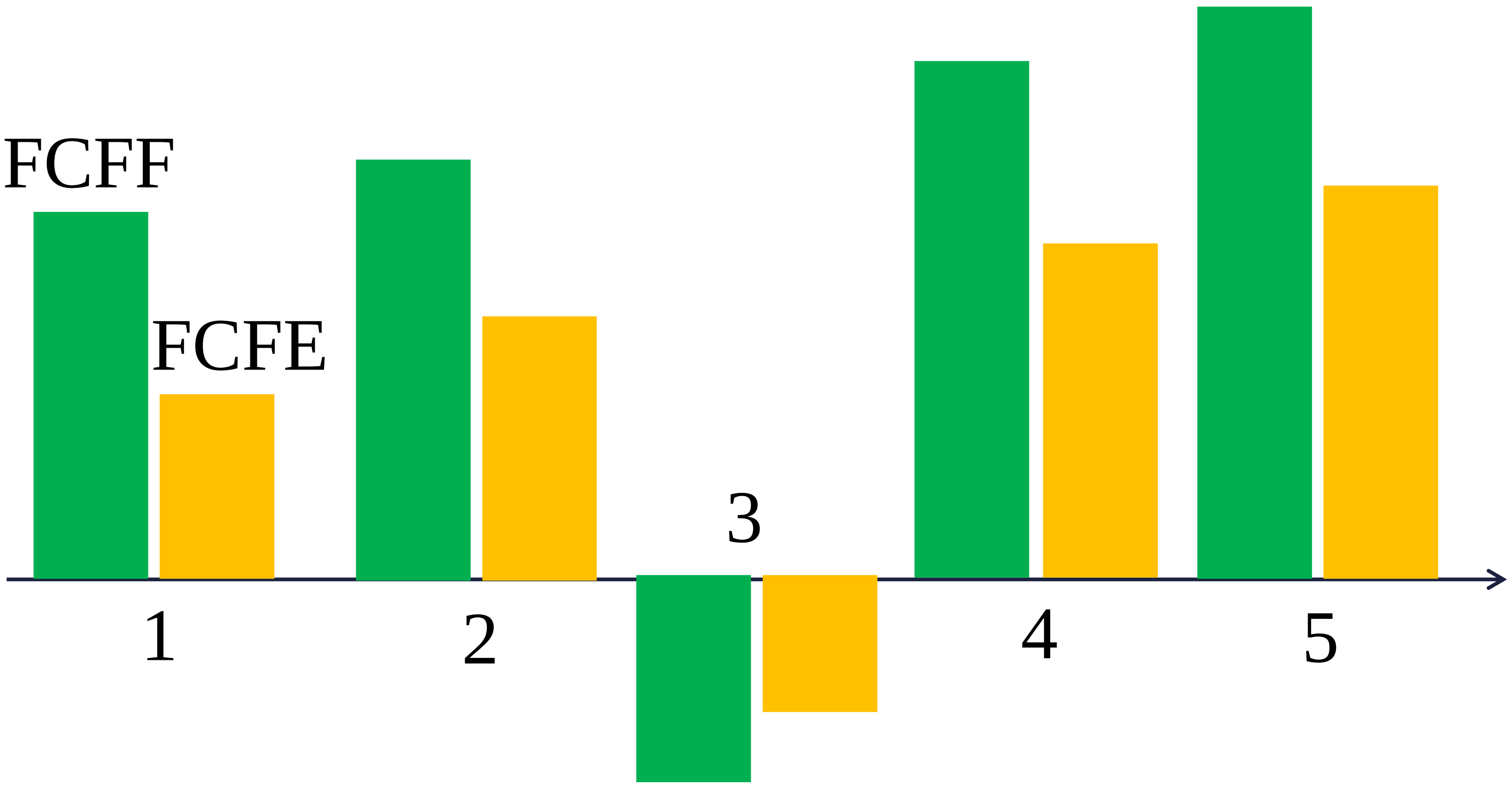
What kind of DCF plan is this? 1



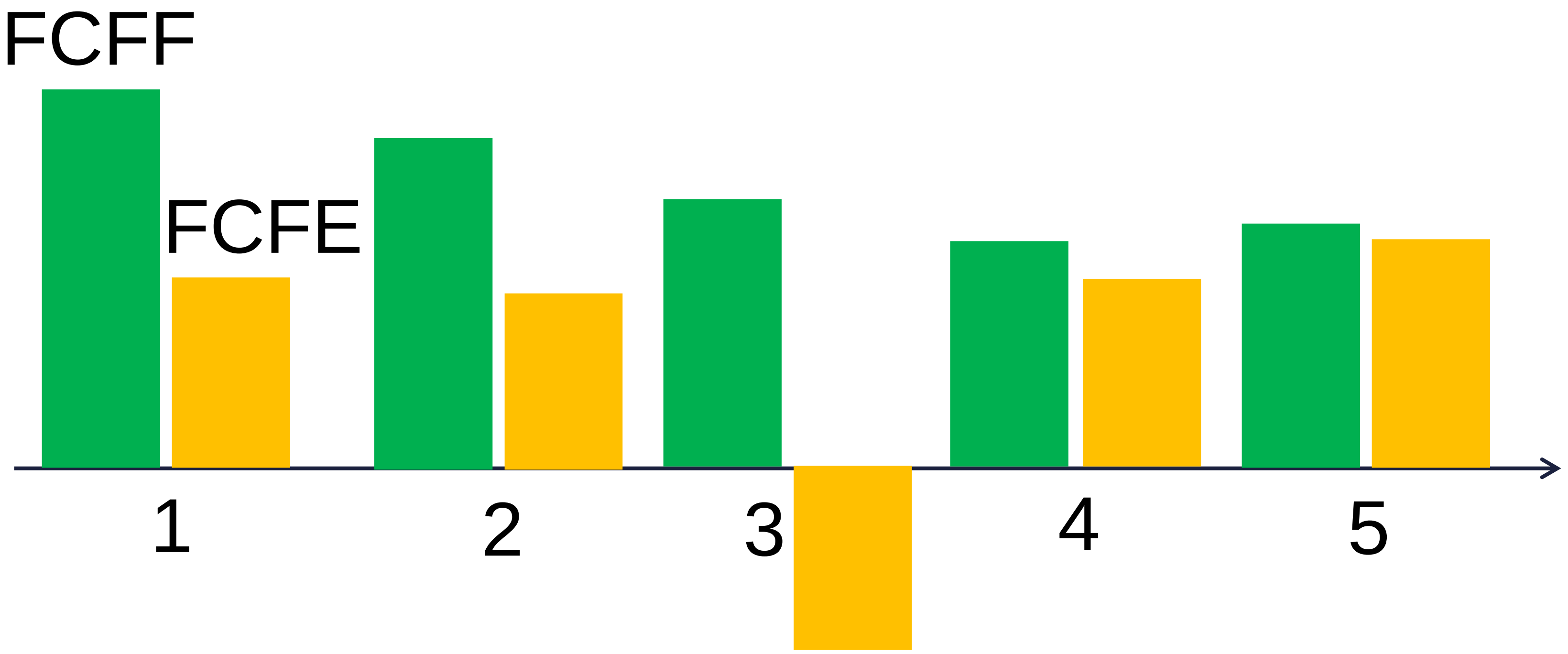
What kind of DCF plan is this? 2



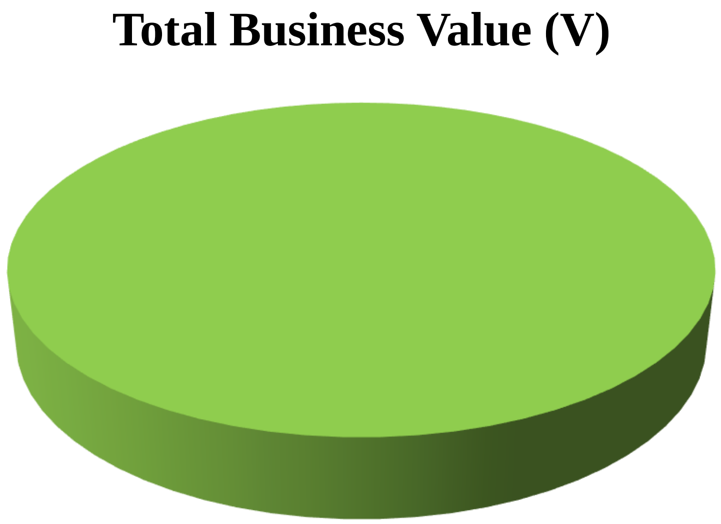
What kind of DCF plan is this? 3

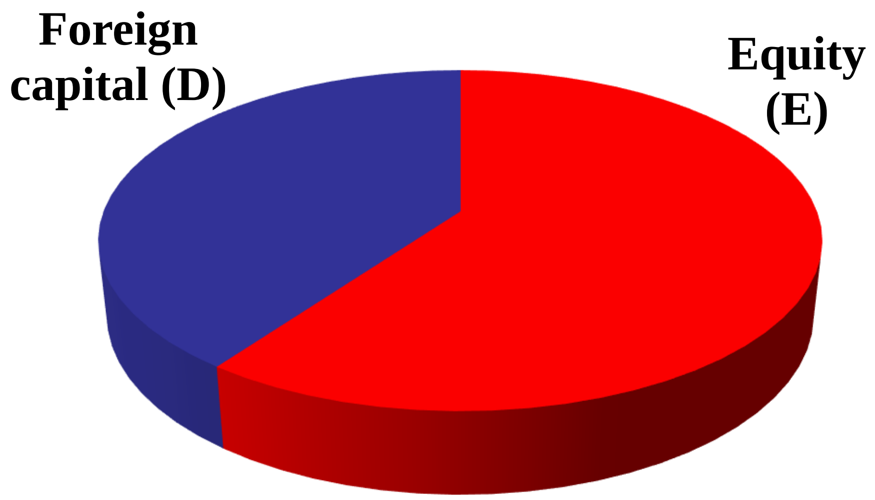


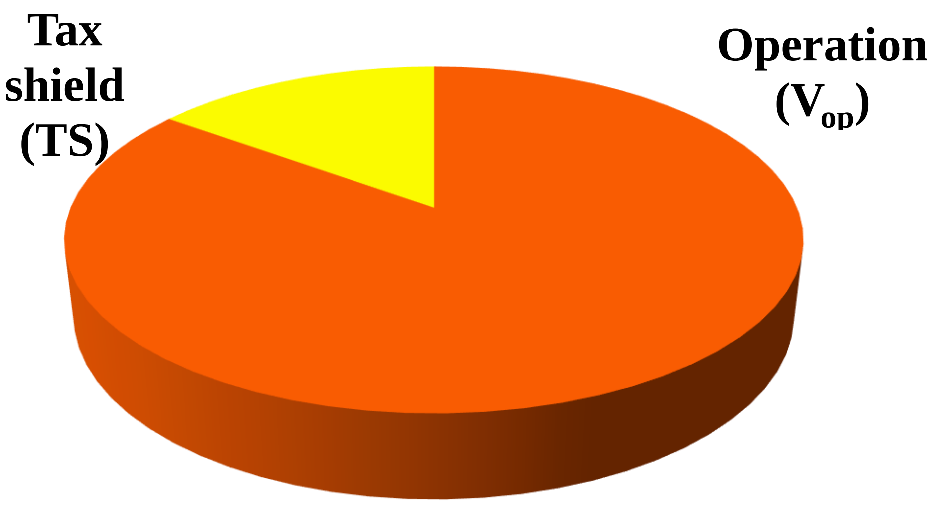
What kind of DCF plan is this? 4

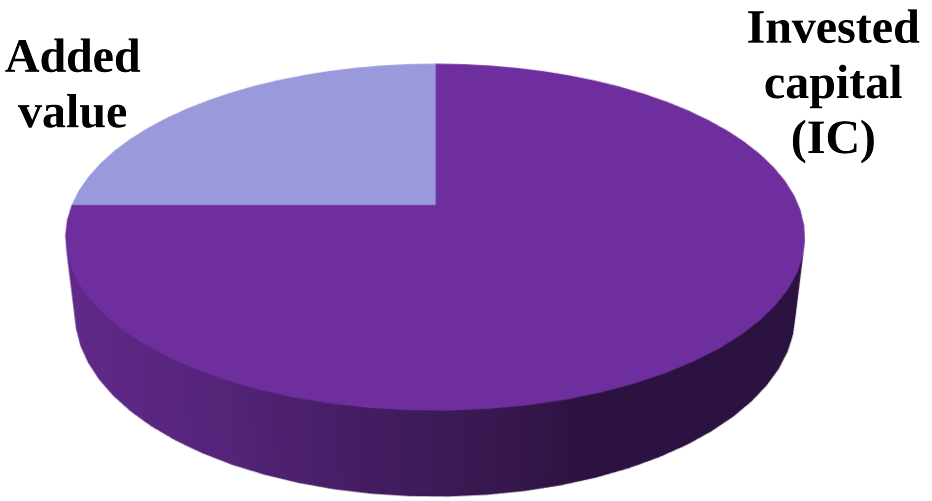


The division of the business value









DCF models

Entity models

- FCFF
- APV
- EVA

Equity models

- Dividend models
- FCFE
- RI (Residual Income)

DCF valuation methods

Entity valuation (V)
where E = V_{op} + NOA - D

Direct Equity models (E)

Dividend Discount Models

FCFE

$$\sum \frac{FCFE_t}{\prod_{i=1}^{t}(1+r_{Ei})}$$

RI

$$\sum \frac{RI_t}{\prod_{i=1}^{t}(1+r_{Ei})} + St_0$$

FCFF

$$\sum \frac{FCFF_t}{\prod_{i=1}^{t}(1+WACC_i)}$$

APV

$$\sum \frac{FCFF_t}{\prod_{i=1}^{t}(1+r_{At})} + PV(TS)$$

EVA

$$\sum \frac{EVA_t}{\prod_{i=1}^{t}(1+WACC_t)} + IC_0$$

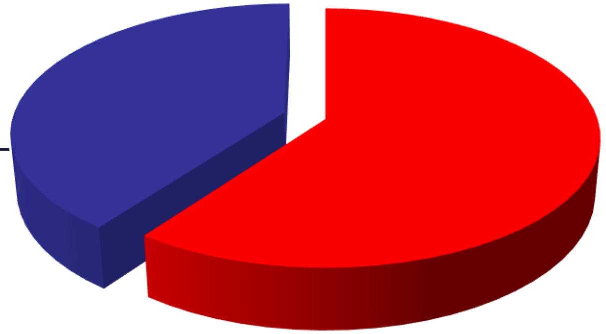
Entity or Equity models?

All DCF models lead to the same result.
(This is the essence of Miller-Modigliani theorems.)

Value of operation (V_{op})
+ Value of non-operative assets

Total firm value (V)
- Value of Foreign capital (D)

= Value of equity (E)



Use rather entity models

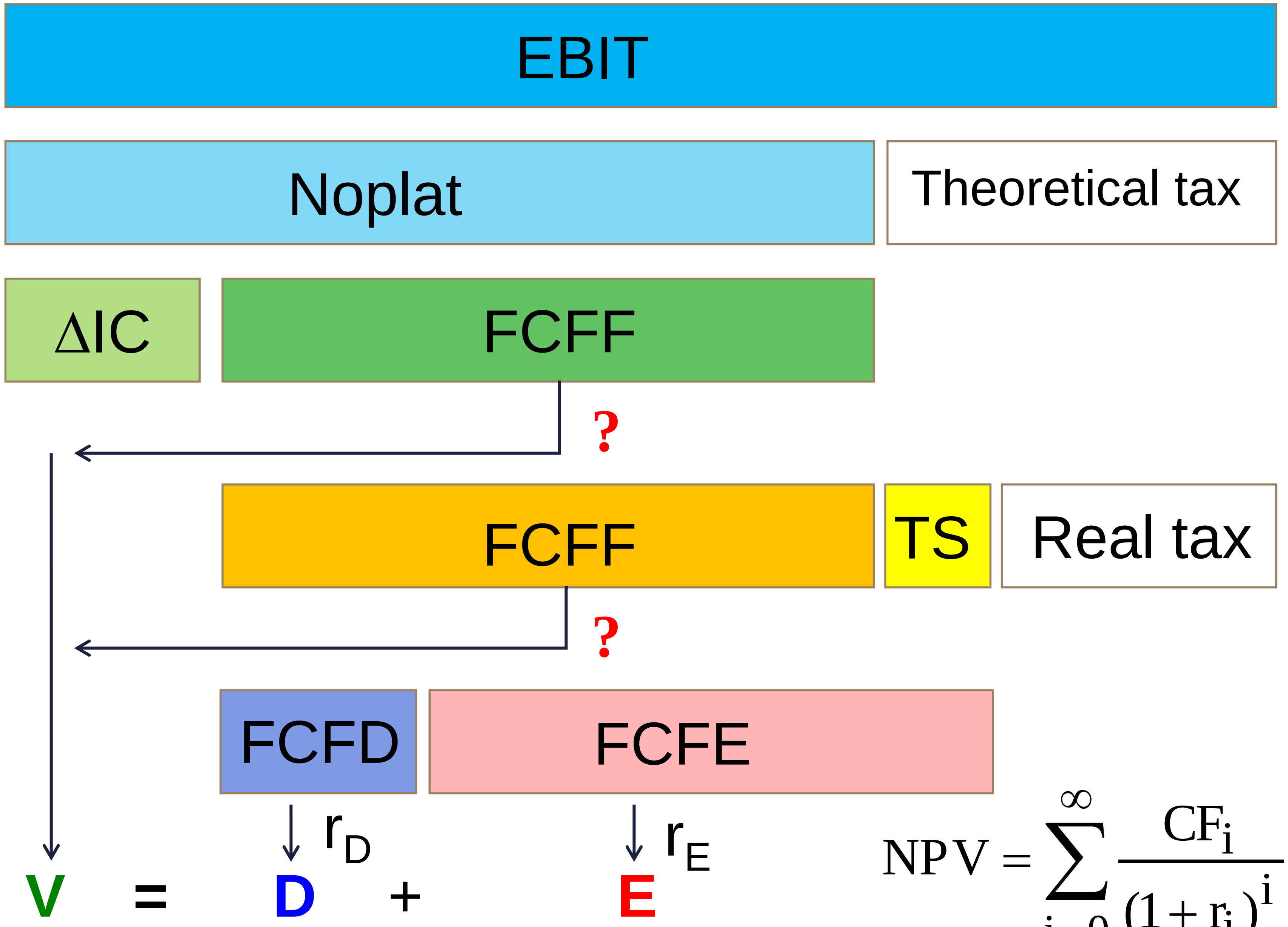
- we usually focus on the value creation process,
- those can be used for Value Based Management (motivation),
- multi-business firms can be analysed,
- offer more freedom to model financing alternatives.

Steps of DCF valuations

1. **Estimating free cash flows**
2. **Estimating cost of capital**
3. Calculating terminal value
4. Calculating current value
5. Sensitivity tests, Scenario analysis
6. Considering other circumstances

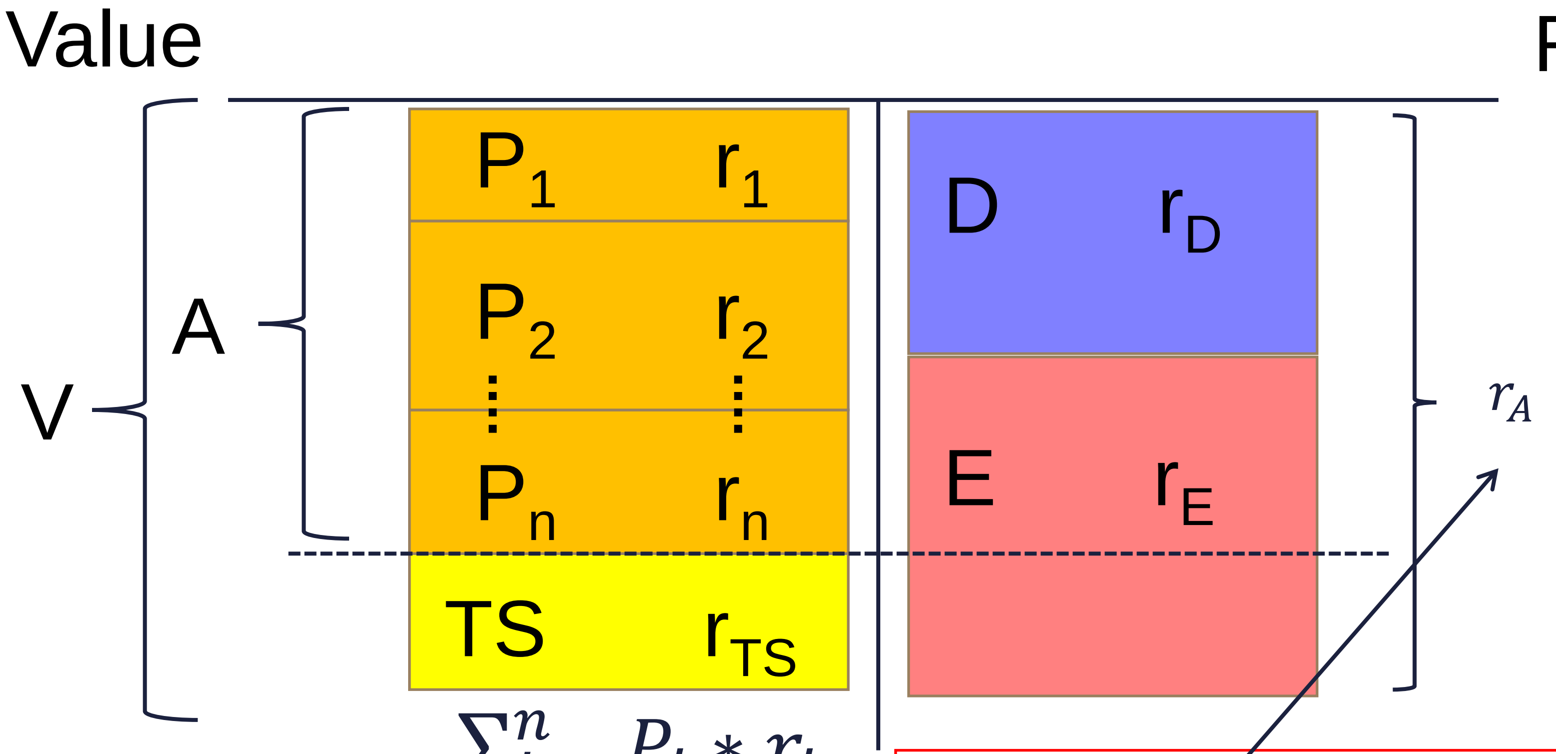
Rules of Cash flow „matching”

- Expiry
- Currency
- Nominal/real
- Effects of financing (TS)
- Risk



$$NPV = \sum_{i=0}^{\infty} \frac{CF_i}{(1+r_i)^i}$$

Operative cost of capital



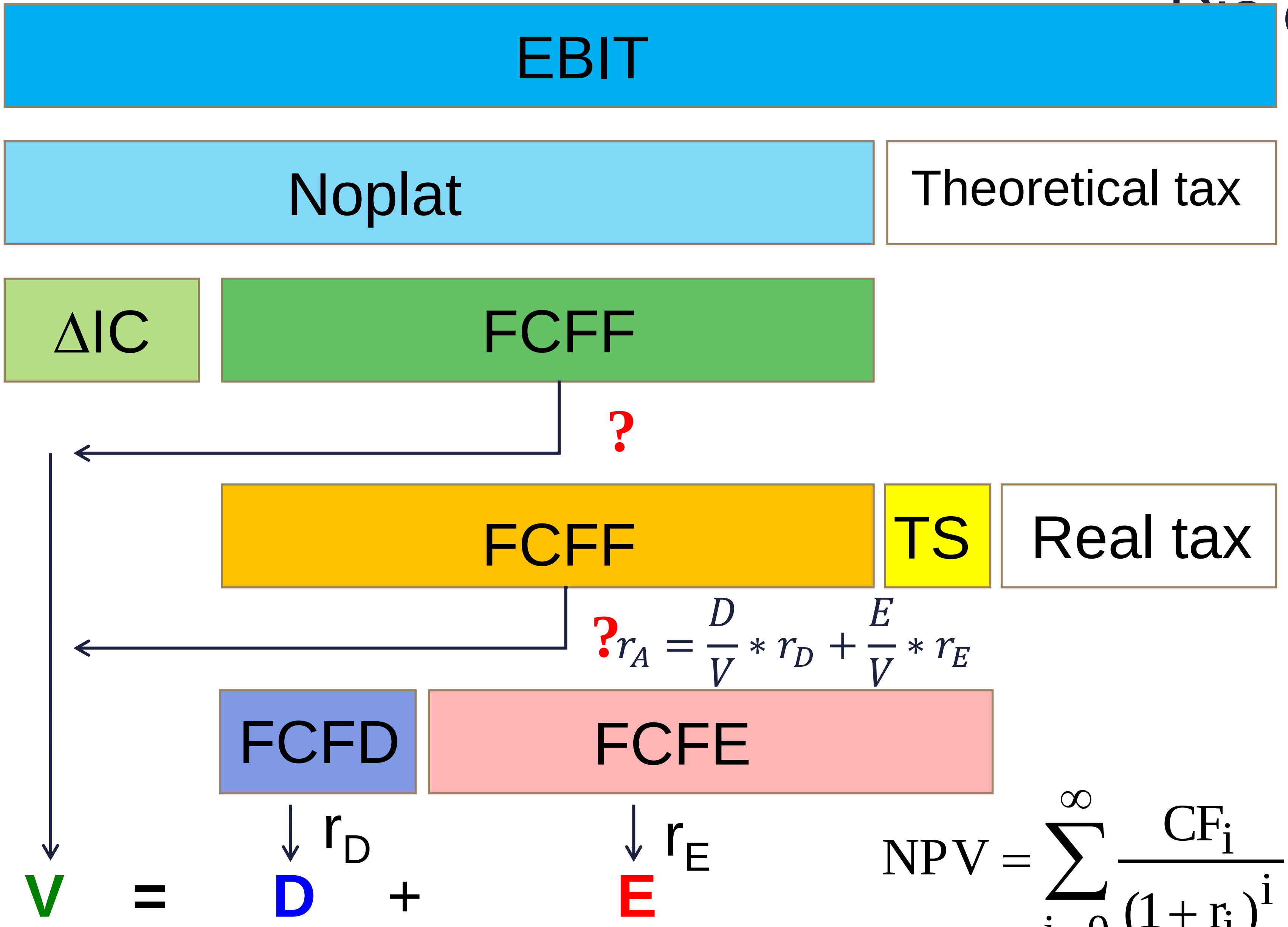
$$r_A = \frac{\sum_{i=1}^{n} P_i * r_i}{A}$$

$$\boxed{r_V = \frac{A}{V} * r_A + \frac{TS}{V} * r_{TS}}$$

If TS=0 or $r_{TS}=r_A$:

$$\boldsymbol{r_V = r_A = \frac{D}{V} * r_D + \frac{E}{V} * r_E}$$

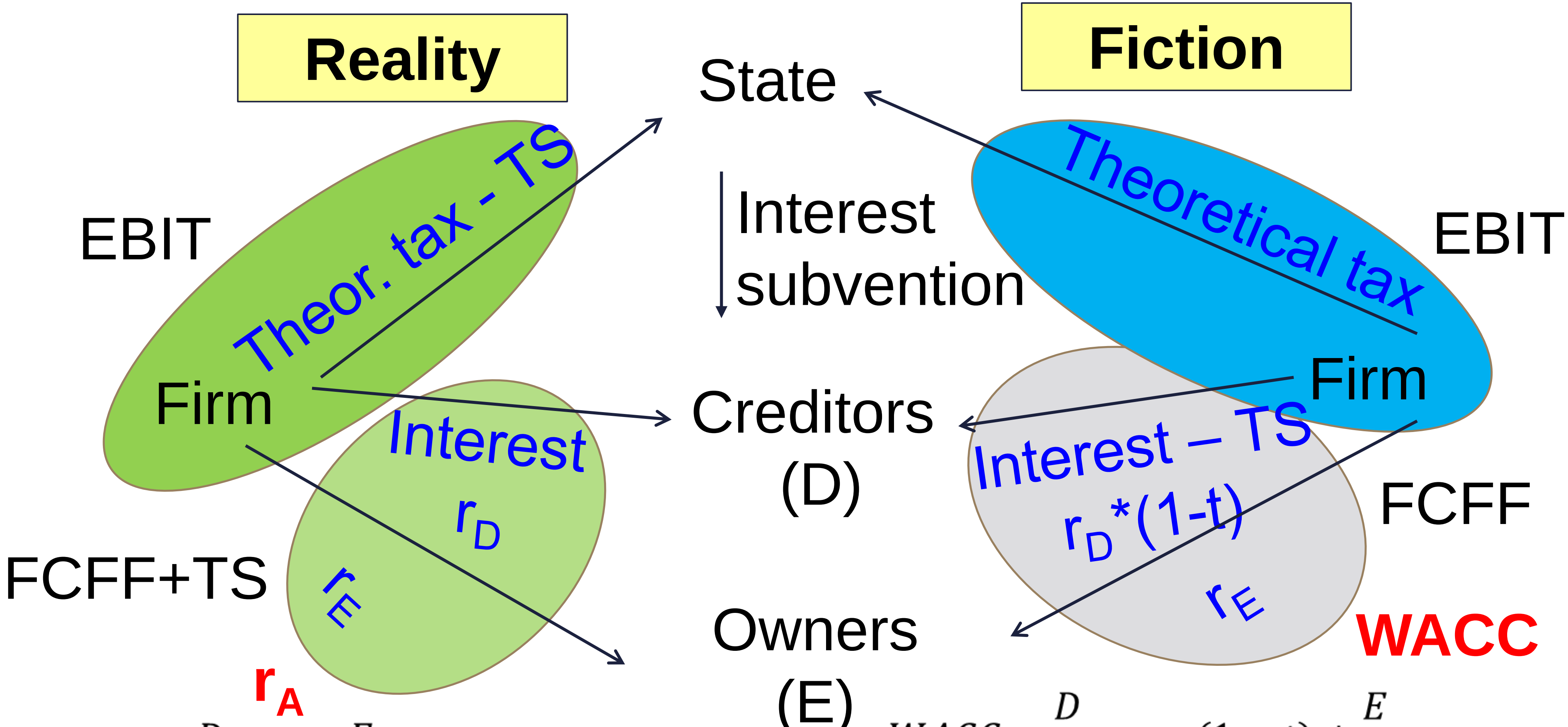
Dia címsor



$$r_A = \frac{D}{V} * r_D + \frac{E}{V} * r_E$$

$$NPV = \sum_{i=0}^{\infty} \frac{CF_i}{(1+r_i)^i}$$

Weighted Average Cost of Capital (WACC)



$$r_A = \frac{D}{V} * r_D + \frac{E}{V} * r_E$$

$$WACC = \frac{D}{V} * r_D * (1-t) + \frac{E}{V} * r_E$$

EBIT

Noplat | Theoretical tax

ΔIC | FCFF

$$WACC = \frac{D}{V} * r_D * (1 - t) + \frac{E}{V} * r_E$$

FCFF | TS | Real tax

$$r_A = \frac{D}{V} * r_D + \frac{E}{V} * r_E$$

FCFD | FCFE

r_D | r_E

V = D + E

$$NPV = \sum_{i=0}^{\infty} \frac{CF_i}{(1 + r_i)^i}$$

FCFF method

$$V = \sum \frac{FCFF_t}{\prod_{i=1}^{t} (1+WACC_i)}$$

FCFF: CF available to all financiers. (CF independent of financing)

V - D = E

FCFE method

$$E = \sum \frac{FCFE_t}{\prod_{i=1}^{t} (1 + r_{Ei})}$$

FCFE: CF available to shareholders. (CF depends on the financing structure, can be redrawn freely.)

E + D = V

Limits of FCFF valuation:

1) shows the value for a given financing structure,

2) financing effects can not be separated.

APV = V assuming equity only financing + Present value of tax shields **[PV (TS)]**

$$V_{Op} = \sum \frac{FCFF_t}{\prod_{i=1}^{t} (1 + r_{Ai})}$$

What rate to use for discounting tax shield?
During this course, we assume tax shield is linked to operation ⇒ $\mathbf{r_A}$

EVA/RI valuation

Limitations of FCFF valuation:

1) The seemingly high proportion of terminal value may create the impression that value is created after the explicit period.

2) does not show when the value is created.

Economic profit = added profit over the required level

$$\mathbf{EVA_i = IC_{i-1} * (ROIC_i - WACC_i)} = NOPLAT_i - IC_{i-1} * WACC_i$$

$$\mathbf{RI_i = BV(E)_{i-1} * (ROE_i - r_{Ei})} = \text{Net Profit} - BV(E)_{i-1} * r_{Ei}$$

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EVA/RI method

$$V = \sum \frac{EVA_t}{\prod_{i=1}^{t} (1 + WACC_i)} + IC_0$$

In the long run ROIC = WACC (ROE = r_E), thus EVA = RI = 0. The added value in TV in such a case is zero.

$$E = \sum \frac{RI_t}{\prod_{i=1}^{t} (1 + r_{Ei})} + BV(E)_0$$

A simple example

What is the value of the firm if it is expected to generate 1500 Noplat next year, ROIC will remain stable at 15%, growth will amount to 5%, operative cost of capital will stay 10%, D/V rate will stick to 50%, r_D=6%, and tax rate is 20%?

Warm-up calculations

Noplat	1500.00
ROIC	15.00%
IC	10000.00

r_A	10.00%
D/V	50.00%
r_D	6.00%
t	20.00%
r_E	14.00%
WACC	9.40%

ROIC=RONIC	15.00%
g	5.00%
K	33.33%
FCFF	1000.00

FCFF method

Noplat	1 500.00
ΔIC	-500.00
FCFF	1 000.00

WACC	9.40%
g	5.00%

V	22 727.27

$$\sum \frac{\mathrm{FCFF}_t}{(1+\mathrm{WACC})^t}$$

$$V = \frac{FCFF}{WACC - g}$$

$$V = \frac{1000}{9.4\% - 5\%}$$

FCFE method

V	22727.27
D	11363.64
FCFF	1 000.00
$D*r_D$	-681.82
$D*r_D*t$	+136.36
ΔD	+568.18
FCFE	1 022.73
r_E	14.00%
E	11363,64
V=D+E	22727.27

$$\sum \frac{FCFE_t}{(1+r_E)^t}$$

$$E = \frac{FCFE}{r_E - g}$$

$$E = \frac{1022,73}{14\% - 5\%}$$

APV method

FCFF	1 000.00
$D*r_D*t$	136.36
r_A	10.00%
g	5.00%

$FCFF//r_A$	20 000.00
$TS//r_A$	2 727.27
V	22 727.27

$$\sum \frac{\mathrm{FCFF}_t}{(1+\mathrm{r}_A)^t} + \mathrm{PV(TS)}$$

$$V = \frac{FCFF}{r_A - g} + \frac{D * r_D * t}{r_A - g}$$

$$V = \frac{1000,00}{10\% - 5\%} + \frac{136,36}{10\% - 5\%}$$

DCF valuation methods

Method	To discount	Discount rate	Notes
Total firm DCF	FCFF	WACC	Useful for multiple business units and set D/V ratio firms
Economic profit	EVA / RI	WACC / r_E	Shows when the value will be created.
APV	FCFF and tax shield (TS)	Operative cost of capital (r_A)	Separates the effect of financing, handy for M&As.
Direct equity valuation	FCFE	Required rate on equity(r_E)	Detailed, hectic financial plan available, bottom-up CF, financial service industry

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Which should I choose?

1. What to do if business units have no stand-alone financing?

2. How risky is this business plan?

3. What to do if it is very hard to estimate the cost of capital for this firm?

4. What is the value of the shares given this outlook?

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Thank you for your attention. Any questions?

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